

SAVING AND BORROWING

Reading Time	150 minutes
Prior Knowledge	None
Keywords	Time Value of Money, Interest Rate, Principal, Time to Maturity, Spot Rate, Forward Rate, Simple Interest, Compound Interest, Yield Curve.

In this lesson, we will examine how you can either earn interest as a depositor or pay interest as a borrower. We'll show how you can calculate the amounts of interest under different types of compounding. We'll also discuss how these interest rates differ regarding how long you're willing to maintain the investment. While the lesson is a bit technical, it only requires a modicum of algebra.

1. Time Value of Money

A very important financial concept is that of **time value of money**. Stated in a rigorous manner, this financial principle says that the value of \$1 today is worth more than the value of \$1 in the future.

There are at least two ways to convince ourselves that the notion makes sense:

- If we received \$1,000 today and we could invest it, say at a 5% rate of return, then in 1 year, we would have \$1,050. Therefore, it seems reasonable that we would prefer \$1,000 today to \$1,000 in 1 year. Alternatively, we would be indifferent toward either \$1,000 today or \$1,050 in one year.
- Another way to think about this concept is to consider that we are willing to forego the utility we derive from spending the \$1,000 today only if we are able to spend more in 1 year's time as economic agents exhibit a preference for instant gratification. In other words, we forego spending in favor of saving or investing that money if we expect that a sufficient increased amount of money will be available to us to spend in one year's time.

2. Interest Rates

Interest rates are a measure of the cost of a loan to a borrower or the compensation we want to receive in order to save or invest a specific amount of money instead of spending it.

When we take out a loan, we must repay the **principal value** of that loan and, in addition, we have to make periodic **interest payments** to the lender. **Interest payments** are specified in terms of an annual rate and depend upon the amount of the loan and its duration. Essentially, the lenders determine interest rates based on the purpose of the loan, market conditions, financial reputation of the borrower, and the amount of the loan.

We should consider that the lender wants to be compensated for giving up immediate consumption, but the lender also wants to be compensated for the risks that the interest payments are not made and/or that the borrower defaults on their obligation to repay the principal. The longer the duration of a loan and the larger the amount of the loan, the riskier it is considered. Economic conditions play an important role too. If, for example, the borrower used the money to start a new business and an economic recession comes along, this can make the success of the new business less likely and the repayment of the loan more uncertain. Furthermore, banks may compete with each other to offer loans to companies and private investors and may offer lower interest rates to attract new business.

3. Mathematics of Present and Forward Values

To be able to perform the correct financial calculations, we must be able to compute the value of investments or financial obligations at different moments in time.

This depends first of all on the way the interest to pay or receive is calculated. There are two types of methods: simple interest and compound interest.

The **simple interest** can be defined as the annual percentage of a loan amount that must be paid to the lender in addition to the principal amount of the loan. The total dollar amount of interest depends on the length of time it takes for the loan to be repaid. The key formula is the following:

$$\text{simple interest} = P \times r \times n$$

where P is the principal amount (amount borrowed), r the annual interest rate, and n the term of the loan in years.

The **compound interest rate** is more complicated because interest is accrued (i.e., accumulated over time) and interest is earned on the principal plus any interest that was paid previously. The key formula is the following:

$$\text{compound interest} = P \times (1 + r)^t - P$$

where P is again the principal amount, r the annual interest rate, and t the number of years applied.

Students should notice that:

- simple interest is interest earned only on the initial investment amount; and
- compound interest is interest earned on the initial investment amount and on interests received. In other words, interest earns interest.

Example. Assume that we can deposit a sum of \$100,000 in bank A that promises to pay us a 5% annualized interest rate using simple interest. Alternatively, we could deposit the same amount of money with bank B, which pays an annual interest rate of 4.5% but using compound interest. Where should we deposit our money if we plan to keep our money as a deposit for 10 years?

Answer. The calculations are simple. If we deposit our \$100,000 with bank A, then after 10 years, we will receive the following dollar amount:

$$100,000 \times 0.05 \times 10 = 50,000.$$

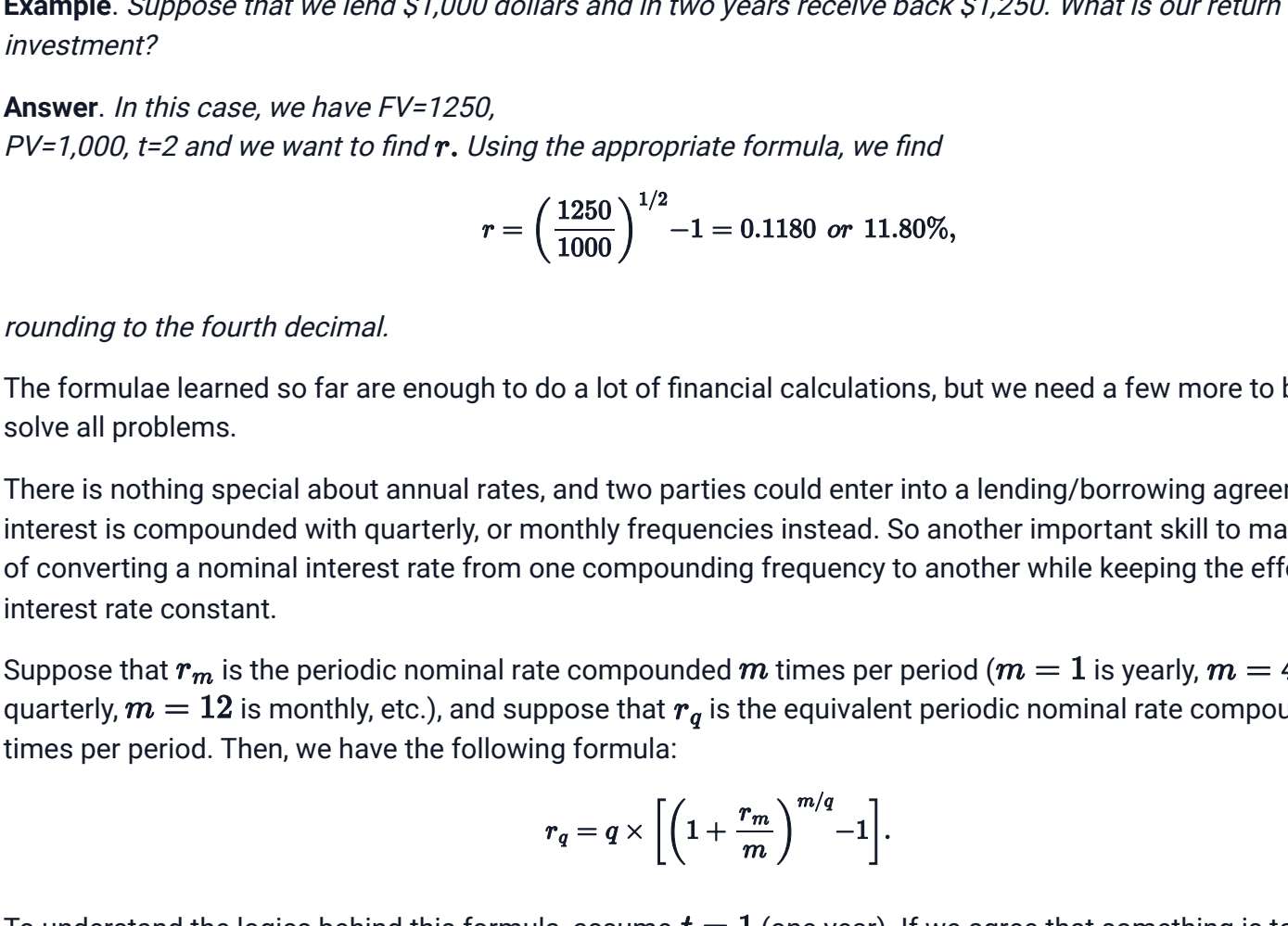
On the other hand, if we deposited our money in bank B, the dollar amount we receive as interest would be

$$100,000 \times (1 + 4.5\%)^{10} - 100,000 = 55,296.94.$$

Hence, despite the fact that bank A pays a higher annualized interest rate, we are better off depositing our money with bank B as the compound interest grows much faster than simple interest over time. On the other hand, if we were borrowing the money, we would prefer to be charged a simple interest rather than a compound interest.

Figure 1: Impact of Compounding Interest Rates

Difference between the simple and compounded interest



If we think of an interest rate as the proportion of a loan that is charged as interest to a borrower, it is the cost of borrowing money (If seen from the point of view of the borrower) or the return on investment (if observed from the point of view of the lender). For the time being, we assume that this interest rate is fixed, but interest rates can be variable as well. For example, the interest on a loan could be paid every 6 months based on the 3-month LIBOR rate for the period that ends 1 month before the payment is due plus 3%. For the time being, we will not address this type of situation.

Example. An economic agent has \$1,000, and she is willing to lend the money to a different economic agent, as long as she is compensated 5% in 1 year. In this case, 5% is the expected rate of return she desires. The borrower agrees to compensate her 5% for 1 year. How much money will she receive in 1 year if the interest rate is compound?

Answer. This is a very simple problem. In fact, the amount she expects to receive back is easily calculated as:

$$1,000 \times (1 + 5\%) = 1,050$$

i.e., the lender expects to receive back \$1,050 in one year.

NOTE. From now on, unless stated otherwise, we assume that interest is compounded when working on applications. The most common type of interest is compound, but there are instances where simple interest is used (e.g., certificates of deposits, among others).

If we'd like to be reasonably formal, we can introduce the notions of present value (PV) and forward value (FV). We also agree to measure time (t) in years, use r for the number of years, and express the desired interest rate/rate of return (r) as an annual percentage. Then, we have the following formulae:

For simple interest rate:

$$FV = PV \times (1 + nr).$$

For compound interest rate, instead, we have:

$$FV = PV \times (1 + r)^t.$$

In the course, we will use mostly compound interest; thus, we will spend a little more time on it. The formula given earlier can also be reversed to yield

$$PV = \frac{FV}{(1 + r)^t},$$

and, when required, it could also be used to recover the required interest rate given PV and FV as:

$$r = \left(\frac{FV}{PV} \right)^{1/t} - 1.$$

Example. Suppose that we lend \$1,000 dollars and in two years receive back \$1,250. What is our return on our investment?

Answer. In this case, we have $FV=1250$,

$PV=1,000$, $t=2$ and we want to find r . Using the appropriate formula, we find

$$r = \left(\frac{1250}{1000} \right)^{1/2} - 1 = 0.1180 \text{ or } 11.80\%,$$

rounding to the fourth decimal.

The formulae learned so far are enough to do a lot of financial calculations, but we need a few more to be able to solve all problems.

There is nothing special about annual rates, and two parties could enter into a lending/borrowing agreement where interest is compounded with quarterly, or monthly frequencies instead. So another important skill to master is that of converting a nominal interest rate from one compounding frequency to another while keeping the effective interest rate constant.

Suppose that r_m is the periodic nominal rate compounded m times per period ($m = 1$ is yearly, $m = 4$ is quarterly, $m = 12$ is monthly, etc.), and suppose that r_q is the equivalent periodic nominal rate compounded q times per period. Then, we have the following formula:

$$r_q = q \times \left[\left(1 + \frac{r_m}{m} \right)^{m/q} - 1 \right].$$

To understand the logics behind this formula, assume $t = 1$ (one year). If we agree that something is to be compounded m times a year then the future value would be:

$$FV_m = \left(1 + \frac{r_m}{m} \right)^m.$$

If we compound it q times a year, then the future value would be:

$$FV_q = \left(1 + \frac{r_q}{q} \right)^q.$$

Now we want to find r_q so that $FV_m = FV_q$; otherwise someone would have an advantage and the other party a disadvantage. This means that we want to solve the following equation:

$$\left(1 + \frac{r_q}{q} \right)^q = \left(1 + \frac{r_m}{m} \right)^m$$

with respect to r_q . Let's take the power of $1/q$ on both sides. This yields:

$$1 + \frac{r_q}{q} = \left(1 + \frac{r_m}{m} \right)^{m/q}$$

and now with some simple algebra we find the formula given above.

Example. Suppose that we deposit a sum of money at the local bank that pays us an annual interest rate of 5% that compounds monthly, i.e., $m = 12$. Nevertheless, we have agreed with the bank that payments will be made quarterly, i.e., $q = 4$, payments. What is the equivalent annual rate that coincides with quarterly compounding?

Answer. In this case, $m = 12$, $r_m = r_{12} = 0.05$, $q = 4$, $r_q = r_4$. Using the formula we wrote earlier, the calculation is simple:

$$r_4 = 4 \times \left[\left(1 + \frac{0.05}{12} \right)^{12/4} - 1 \right] = 0.050209 \text{ or } 5.0209\%.$$

Let's check our calculations. Assume that we deposited \$10,000. If the interest rate were compounded monthly and credited to our account monthly, then after one year, the value would be:

$$FV = 10,000 \times (1 + 5\%/12)^{12} = 10,511.62.$$

On the other hand, if the interest rate is compounded and paid quarterly, then we find:

$$FV = 10,000 \times (1 + 5\%/4)^4 = 10,509.45,$$

as expected.

It is possible to compound the interest yearly, semi-yearly, quarterly, monthly, daily, every second, and ideally in a continuous manner. Assume that $m = 1$ so that $r_m = r$ is the yearly interest. And we assume that we compound the interest q times a year. If we start with a principal, PV , then the forward value, FV , over a period of length t , assuming that we compound the interests q times, is given by:

$$FV = PV \left(1 + \frac{r}{q} \right)^{qt}.$$

Now, if we take the natural log on both sides we get

$$\ln(FV) = \ln(PV) + qt \ln \left(1 + \frac{r}{q} \right).$$

We want to see what happens when $q \rightarrow \infty$, i.e. we compound interest with an infinite frequency. It is easy to see that

$$\lim_{q \rightarrow \infty} \frac{\ln(1 + r/q)}{r/q} = 1$$

and this is the key for our derivation. In fact, we have

$$\lim_{q \rightarrow \infty} \ln(FV) = \ln(PV) + \lim_{q \rightarrow \infty} qt \cdot \ln \left(1 + \frac{r}{q} \right)$$

or, with some routine algebraic manipulations,

$$\lim_{q \rightarrow \infty} \ln(FV) = \ln(PV) + \lim_{q \rightarrow \infty} tr \cdot \frac{\ln(1 + r/q)}{r/q} = \ln(PV) + tr.$$

Thus we have shown that (using properties of logarithms):

$$\lim_{q \rightarrow \infty} \ln(FV) = \ln(PV) + tr = \ln(PV) + \ln(e^{tr}) = \ln(PVe^{tr}).$$

Hence,

$$\lim_{q \rightarrow \infty} FV = PVe^{tr}$$

Example. Consider an investment of \$10,000 that draws an interest rate of 6% a year continuously compounded. How many years does it take to double our investment?

Answer. Using the formula above, we need to solve the following equation:

$$20,000 = 10,000 \times e^{0.06t}$$

with respect to t . This equation can now be written as

$$2 = e^{0.06t}$$

and taking the natural log on both sides,

$$\ln(2) = 0.06 \times t$$

or

$$t = \frac{\ln(2)}{0.06} = 11.5525.$$

Hence, it takes a little longer than 11.5 years to double our investment. Note that the result is independent from the amount invested.

Compounding is an amazing thing. In fact, Albert Einstein himself is reported to have said "Compound interest is the eighth wonder of the world. He who understands it, earns it, ..., he who does not pays it." See "10 Reasons Why Compounding Interest Is the 8th Wonder of the World" for some optional reading on that topic.

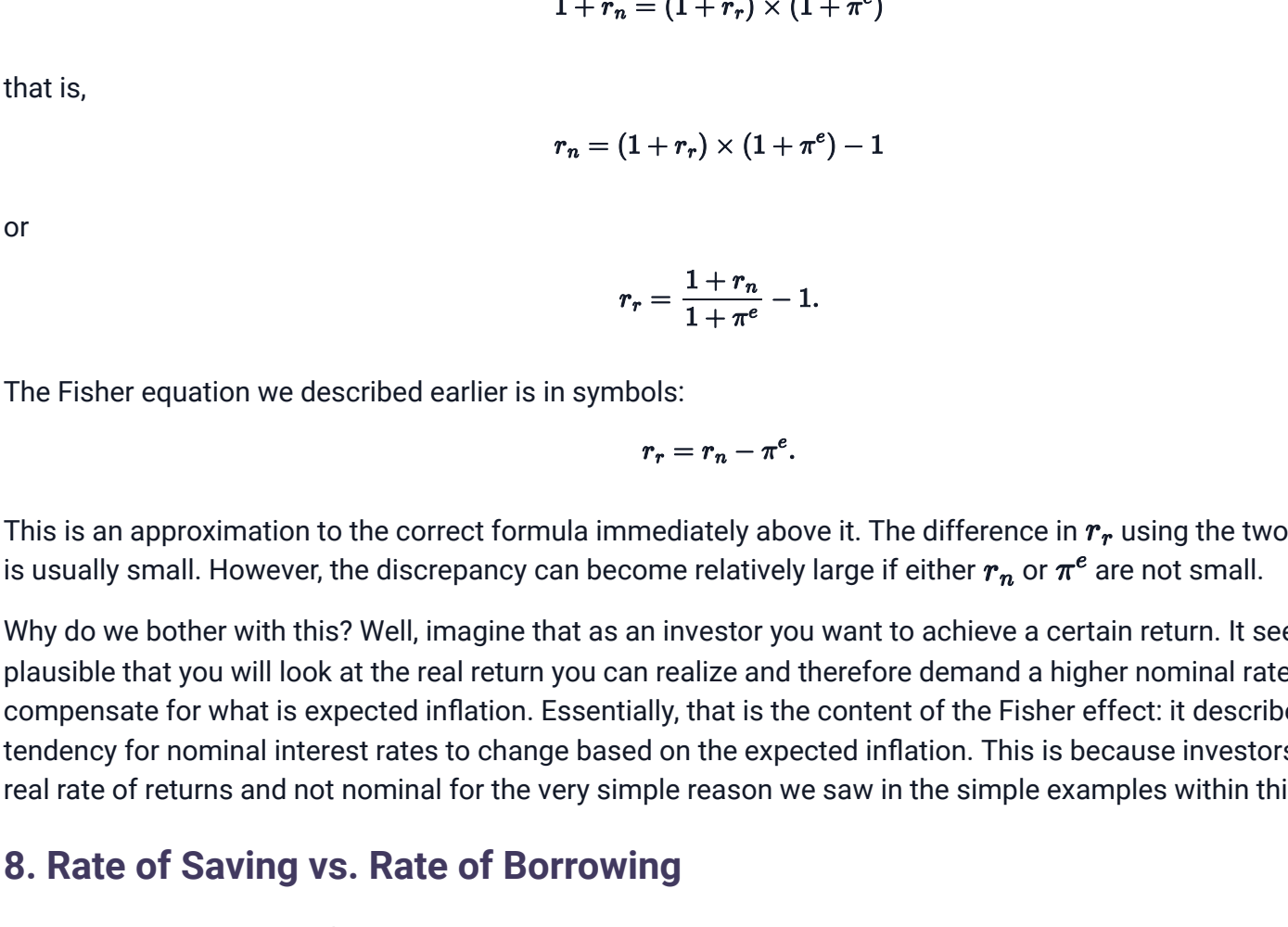
Students can also utilize this [Shiny application](#) to comprehend the exponential nature of compound interest. You will be able to visualize the growth of investments over time, attributable to the compounding effect. It is interesting how the nearly linear growth observed with low interest rates and fewer compounding periods transitions to an exponential growth pattern when higher interest rates, more frequent compounding periods, and extended time frames are applied.

4. Yield Curve

Short-term interest rates are usually lower than long-term interest rates because typically long-term investments are riskier and investors want to be compensated to take the additional risk. However, this is not always so, and in some periods, the opposite could be true. A good indication of the difference in rates between short- and long-term investments can be investigated via the **yield curve**.

The yield curve refers to the relationship between the short- and long-term interest rates of fixed income securities (bonds) issued by the U.S. Treasury. In Figure 2, we can look at three different yield curves that were available on three different dates.

Figure 2: Three Snapshots of the Yield Curve at Different Dates



As mentioned at the beginning of this section, the yield curve typically has a positive slope. In some cases, we may observe what is called an **inverted yield curve**, in which short-term interest rates are higher than long-term interest rates. An inverted yield curve is telling us that near-term investments are riskier than the long-term ones. Often, an inverted (Treasury) yield curve is a leading indicator of the approaching of an economic recession (for those with no background in Economics, imagine that the government wants to stimulate the economy to counter a recession and one way to do this is to lower interest rates; that is the cost of money for those who are willing to invest).

5. Spot and Forward Rates

To complete this lesson, we'll use an example to illustrate another important issue that is relevant in a number of practical problems. Before doing that, we'll introduce two important concepts.

A **spot interest rate** is the current rate at which we can invest our money. We may be told that if we invest for 1 year, the annual interest rate is, for instance, 5%; if we invest for 2 years, the annual rate of interest rate is 5.5%, and if we invest for 3 years, then the annual interest rate is going to be 5.8%. This is in line with what we said earlier about longer durations requiring higher interest rates. There are all examples of spot rates for different maturities of the proposed investment as they are known right now at time 0.

However, while we may be interested in investing for just 2 years, we are not willing to invest our money immediately. We want to invest for two years starting at the end of the first year. So we cannot assume that we will receive 5.5%. We want to find out what interest rate we are going to earn on such an investment in order to make comparisons. This leads us to deal with the concept of **forward interest rate** when dealing with an investment that will not take place until a predetermined period.

For this topic, notation is very important to avoid confusion, so we will use heavy notation that does not leave much room for guessing. Suppose that $B(0, t)$ is the price (market price, as we assume that the bonds are traded) at time 0 of a bond that pays \$1 at time t . Similarly, we use $\tau(0, t)$ to denote the annual interest rate for an investment in which we enter today and that matures at time t . Using the principle of compound interest, we find:

$$B(0, t) = \frac{1}{(1 + \tau(0, t))^t}$$

If we could either solve this equation or, perhaps, we can get $\tau(0, t)$ from a yield curve.

If $\tau(0, t)$ is the rate at which we can invest today in a t period bond, we can define an implied forward rate between the years $s \geq 0$ and t with maturity $t > s$ as the interest rate that applied to an investment we enter into at time s and that last $t - s$, i.e., reaches maturity at time t . We indicate this implied forward rate using the notation $f(0, s, t - s)$. Then, the formula for the forward rate in terms of the spot rates is given by:

$$f(0, s, t - s) = \left[\frac{(1 + \tau(0, t))^t}{(1 + \tau(0, s))^s} \right]^{1/(t-s)} - 1,$$

and $f(0, s, t - s)$ is the forward rate as seen from date 0, starting at time s and with maturity at time t in $t - s$ years.

Going back to our example, we should be able to recognize immediately that for us $\tau(0, 1) = 5\%$, $\tau(0, 2) = 5.5\%$, and $\tau(0, 3) = 5.8\%$.

Then, the formula just introduced yields

$$f(0, 1, 1) = \left[\frac{(1 + 5.8\%)^2}{(1 + 5\%)^1} \right]^{1/1} - 1 = 6.00\%.$$

$$f(0, 1, 2) = \left[\frac{(1 + 5.8\%)^3}{(1 + 5\%)^1} \right]^{1/2} - 1 = 6.202\%.$$

$f(0, 1, 2)$ is the interest rate that we were looking for for our investment.

Hence, if we invest for 2 years, at the end of year 1, or equivalently at the beginning of year 2, given the structure of spot rates, we are going to earn an annual return of 6.202%.

Example. In the same setup of the example we just discussed, compute $f(0, 2, 1)$.

Answer. Note the following: here $s = 2$, $t - s = 1$, and therefore $t = 3$. Then, using the formula for the implied forward interest rate, we find

$$f(0, 2, 1) = \left[\frac{(1 + 5.8\%)^3}{(1 + 5.5\%)^2} \right]^{1/1} - 1 = 6.403\%.$$

We are not going to insist on this, but as general knowledge, the ambitious student should be able to recognize that the following formula holds:

$$\tau(0, t) = (1 + \tau(0, 1))(1 + f(0, 1, 1))(1 + f(0, 2, 1)) \cdots (1 + f(0, t - 1, 1))^{1/t} - 1,$$

and that the formula can be used to derive implied forward rates.

6. Variable Interest Rates

Suppose now that your banker suggests that you invest in a variable interest rate. More specifically, the banker suggests that you invest in a variable-rate certificate of deposit (also called a flex CD). A variable rate CD allows you to put your money into a secure protected account (in the U.S., they are protected by the FDIC up to \$250,000) where it will earn interest over the life of its terms (typically 12, 24, or 36 months). A flex CD pays an interest rate that can go up and down during the life of such security (the exact factors that determine the actual interest rates are not important right now).

Example. The banker suggests that we invest in a 24-month flex CD, which offers a rate based on the federal funds rate minus 0.3%. Currently, the federal funds rate is 3%, and we decide to open this account. Then, our flex CD account starts earning 2.7%. After one month, the federal funds rate goes up to 3.5%. Hence, now, our flex CD account earns 3.2%. Assume that the federal funds rate continues to increase, and after 6 months, it is 4%. Then, after 6 months, our flex CD account earns 3.7%. Going forward, after 6 more months, the federal funds rate goes to 3.8%. Then, at this point, our account earns 3.5%, and so on.

Let's consider a few facts. While CDs usually pay monthly compounded interest, we will use for simplicity that they pay simple interest. Then, assuming an investment of \$1,000, each month we receive:

$$\text{simple interest} = 1,000 \times ? \times 1/12$$

where here ? indicates that we do not know the unknown rate to be used.

Since the rate is unknown, we cannot calculate with certainty what the earned interest will be. We cannot calculate for sure how much we will have in three months, or six months, or 24 months. However, we can estimate using some rates. If you think interest rates will increase, you may opt for a variable rate to take advantage of higher interest rates leading to more interest. On the other hand, if you think interest rates will decrease, you may opt for a fixed rate to lock in a higher rate. Note that even for a simple loan, there are many decisions that must be made:

- The amount of principal to deposit
- The amount of time to "lock up" the money
- The choice of a fixed or variable rate

Mortgages can also be had with fixed or variable rates. In this case, the opposite is true if you ask the bank for a mortgage, and we would prefer the option that allows us to pay the least each month.

In either case, these decisions may not be as simple as they seem.

First, do we know how much we can lock up right now? What do we know about our future income and expenses?

Second, are we sure that the amount of time the money is locked up for is reasonable? Will we need the funds sooner? In the case of CDs, for instance, there might be harsh penalties for withdrawing the money before maturity.

Third, we can anticipate rates going higher (or lower), but what if we are wrong?

Certainly, these decisions are related. Would I be willing to take a lower fixed rate for a shorter amount of time? Or could I split my deposit across different times so that I can take advantage of not only some shorter terms but also some higher rates?

Much of finance involves *making decisions under uncertainty*. Indeed, this will be a theme we will see well beyond this course and in the very heart of financial engineering itself

Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation should take you directly to the reading or to a page where you can download it.

1. Schenk, Catherine R. "Summer in the City: Banking Failures of 1974 and the Development of International Banking Supervision." *The English Historical Review*, vol. 129, no. 540, 2014, 1129–1156, <https://doi.org/10.1093/ehr/ceu261>

- **Required pages:** 1129–1131
- **Estimated reading time:** 30 minutes

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

COUNTERPARTIES AND CREDIT RISK

Reading Time	90 minutes
Prior Knowledge	Interest
Keywords	Credit Risk, Collateral, Regulation, Deposit Insurance, Bonds

In the previous lesson, we looked at time value of money and the mathematics of present and forward value together with some elements of interest rate theory. The knowledge of these concepts are essential for analyzing the interaction between a borrower and a lender, but they are also powerful tools that allow us to solve important practical problems. One of the justifications for interest rates usually being higher for longer-duration investments has to do with the risk that the borrower may default.

In this lesson, we examine the process of default and the broader picture of credit risk. We will examine how institutions, regulation, and insurance all help to keep this credit risk in check.

1. Risks in Finance

The world of finance is full of risks. We cannot avoid risks altogether because like we sometimes hear “*no pain, no gain*.” In fact, a big part of what a quantitative analyst does deals with recognizing risks, measuring them, and hedging them.

Risk can be divided according to two main categories: **systematic** and **non-systematic**. Systematic risk affects the entire market while non-systematic risk is asset-specific. During your time in the WQU program, you will learn a great deal about risks and how to handle them.

Right now, we will only mention a few main types of risk:

- **Economic risk:** the risk that adverse changes in economic conditions negatively impact businesses, industries, or even entire economies. As investors, we must pay close attention to the economic conditions of those countries in which we plan to invest.
- **Equity risk:** the risk that stock prices will change in an unfavorable manner. If we are investors buying equities, we do so hoping to earn a return that exceeds the risk-free interest rate or else we would not be compensated for taking the risk.
- **Interest rate risk:** the risk that interest rates can change in an unfavorable manner. Imagine a bank that provides long-term mortgages, say 30-year mortgages, and assume that it does so at 5%. Suddenly, interest rates go up to 8%. In this case, the consumer is happy, but the bank not so much. Alternatively, suppose that you have bought a bond that pays an annual coupon of 4.5% and the newly issued bonds pay a coupon of 6.4%. Now, as an investor, you would not be happy, and if you were to sell your bonds, you would have to sell them below par value, which would lead to capital loss.
- **Model risk:** the risk that we use the wrong model to analyze a specific investment. If the model is wrong, it is quite doubtful that our practical solutions are going to be optimal. As quantitative analysts, we must always consider this risk and make sure we employ reliable modeling techniques while being on the lookout for what could go wrong.

The four risks we have listed above are just a few among many more that are possible. In the next sections, we will consider **credit risk** specifically.

2. Defaulting

Before we consider credit risk, we need to understand the concept of defaulting, i.e. not paying in full when a payment is due. In the previous lesson, we realized that not all borrowers are necessarily able to pay their debt. Before a true case of defaulting occurs, it is possible that we run into an **overdue payment** issue, i.e., a borrower fails to make a timely payment of interest, principal, or even both.

It could be a simple case in which the borrower simply pays late. An overdue payment is considered to be any payment that is past a clearly defined and agreeable payment date. Sometimes, late payments occur due to holidays or weekends delaying the processing of payments. There's always the excuse that the payment is lost in the mail. But a payment is overdue because someone simply did not make the payment.

So, what happens when a borrower doesn't pay? Depending on the type of loan, there could be fees that are assessed. Let's look at both sides: from the depositor's perspective and from the perspective of a lender. First, let's consider the depositor's perspective. After the term to maturity, the depositor is expecting to receive both the earned interest and the principal back. For some reason, suppose the bank failed to pay this. Most likely, this would be a bank error and could be fixed quickly. It's unlikely the bank will have to pay more because the fine print that is part of the agreement may not hold them liable for such operational mishaps (operational risk). Nevertheless, banks invest huge amounts of capital to build computational infrastructure to handle the automatic crediting of interest, so these errors are not very common.

Now let's suppose that borrowers are the ones failing to pay on time. Of course, this can be for any number of reasons. In this case, the banks have **grace periods**, which are extra amounts of time to facilitate the payment being processed. If the payment is still not made by the grace period, then there might be extra fees the bank charges, such as late fees, as well as even raising the interest rate for non-payment. Overdue payments are serious and can be the precursor to an actual default. Let us turn to default, the topic of our next section.

Defaulting is much more serious than just being late on a payment. What does it mean to default? Well, borrowing funds has the implied promise that the funds will be returned in full. In addition to the interest payments, the principal repayment is expected. Default occurs when the borrower fails to make some combination of interest payments, principal payments, or both. A default is an overdue payment that remains unpaid. There may even be multiple payments in default. This is particularly important for commercial businesses that may have several debts. Those debts are associated with what are called **financial covenants**. Debt covenants are essentially restrictions that lenders impose on debt holders and that appear in writing in the lending agreements to ensure that a company will operate within certain rules set by the lenders. Lenders do so to avoid situations where a company may be too aggressive in borrowing money and fail to repay them. Often, a default on one loan can trigger a default on all loans with the requests to pay back all of them. When this happens, unless the company has sufficient liquidity, the request could lead to bankruptcy. Hence, defaulting is not to be taken lightly.

We have established that a default occurs when the borrower fails to make payments (interest or principal) to the lender, i.e., the borrower is not owing up to the responsibility that was set out in the terms of the loan. Clearly, this means that the lender is not going to receive their money in full. It is certainly possible that they may receive a partial amount. Can the lender sue the borrower for the balance? Well, virtually every country has laws that protect borrowers who are unable to make full payments by their declaring a type of **bankruptcy**.

Interestingly, the word bankruptcy comes from the Italian term, *banca rotta*, which means "broken bench." At one point in history, these money dealers worked from benches. If a money dealer ran out of money, they would be taken out of business by having their bench broken. With a *banca rotta*, the dealer would no longer be able to conduct business.

So, the bankruptcy/"broken bench" notion is meant to convey that the person is out of business. Fortunately, bankruptcy laws today protect borrowers, which could include small business owners. They may be allowed to continue operating their business, and the amount they are legally required to pay their lenders may be reduced.

While defaulters are generally able to avoid "debtor's prison" (which was also common in history) or serious criminal penalties, they do suffer a financial setback in terms of their credit becoming less favorable.

For example, suppose you are a lender. You are looking to lend money to one of two different people. The first person has an excellent history of making payments. The second person, however, has consistently declared bankruptcy a number of times. Which of the two borrowers do you think would be given a better interest rate? Well, from a bank's perspective, the borrower who defaults poses such uncertainty as to whether the bank receives its funds back. The bank may require a much higher interest rate to offset that greater uncertainty. This is essentially the first real challenge we face as financial practitioners: the challenge of credit risk.

3. Credit Risk

In this section, we introduce credit risk. **Credit risk** is the risk to the lender that the lender does not receive the full amount of principal and interest payments initially spelled out in the loan. Credit risk is a risk that there will be an expected loss due to the inability or even unwillingness of the borrower to repay their obligations. Credit risk tends to affect the lender and not so much the borrower.

Example. *An investor buys 10,000 shares of company X at a price of \$15 each. Six months later, the price of company X's shares is down to \$12 a share. Furthermore, company X decides not to pay dividends for the foreseeable future. A student reported that this happens because the investor had credit risk since company X fails to pay dividends and did badly considering that its stock price went down. Is the student correct?*

Solution. *No. First of all, dividends are at the discretion of company X and buying shares is nothing like buying company X's bonds where coupon payments are required if promised. Furthermore, company X never promised to guarantee the value of the shareholders' investments in its equity, again unlike bonds, which guarantee that the full principal amount will be repaid.*

Based on the example just discussed, we should note that credit risk requires forms of contractual obligations and should not be confused with market risk or other types of risk.

Now, we will illustrate a new and important concept. Suppose you put \$10,000 in a savings account. You wonder if that money will be there when you wish to withdraw it. You also wonder if the bank will honor the interest payments you are meant to receive. Indeed, there have been times in history where banks have had **bank runs** and were not able to supply the depositors with their own money. Please refer to the required reading for this lesson, which discusses a bank failure nearly half a century ago and how it continues to affect supervision in banks today.

Well, banks do not just keep the money locked in accounts; they need to use it for other purposes. Throughout history, the banks may have invested in opportunities that turned out to be deeply unprofitable and lost much of the depositor's funds. Some depositors who lined up early to withdraw may have received their fair share, but others who came later were left with losses, whether through the loss of value or through some operational mishap.

So how can we minimize this credit risk from the depositor's point of view? Well, many countries today are part of the [International Association of Depositors Insurers, or IADI](#). In the United States, this is handled by the FDIC: Federal Depositary Insurance Corporation. [They maintain a directory of other countries in the program](#). In South Korea, this is handled by the KDIC, the Korea Deposit Insurance Corporation.

These corporations basically guarantee that there are mechanisms, with the backing of the federal government, to ensure that depositors will never lose their deposits (up to a certain amount, at least) even if the bank itself goes bankrupt. This is a very important idea and our first major theme in the course of credit risk. In order for counterparties to engage in transactions, there has to be a clear understanding of the risks of each other's credit. Depositing funds in a bank account is truly one of the safest transactions in the world of finance. The credit risk is the risk that the bank would be unable to pay interest and principal back. These Depositary Insurance corporations reduce the risk by providing backup for those banks. These corporations, with the help of government agencies, will require banks to show prudent use of those funds to avoid situations that might cause them to lose the deposited amount. These corporations ensure that there is widespread trust and confidence in the financial system. Can you imagine if people put money in banks, then the banks declare bankruptcy and depositors lose their hard-earned savings? People would be afraid to use banks. Banks are the cornerstones of the financial industry.

Let's switch directions and look at credit risk from the bank's point of view when lending to an individual or small business. Unlike banks, an individual borrower does not have the backing of the Insurance Corporation. Instead, a borrower's creditworthiness comes through their reputation to repay their debts in a timely manner. Nevertheless, an individual may have unexpected income loss due to a job loss or simply unforeseen higher expenses like medical expenses. Any circumstance that hinders the ability of the borrower to repay a debt is an example of credit risk.

Credit risk converts much of finance from deterministic problems to random problems. Let's consider the lender's perspective. The amount of total funds received should be known. Several times, we have seen the equation that shows the amount of interest received. Yet, thanks to credit risk, not all payments are guaranteed to be made. They have some uncertainty. There is a small probability associated with the timing and/or the size of each payment. Credit risk is endemic to virtually all transactions where there are counterparties engaged in a transaction. One thing that can help mitigate credit risk is collateral, the topic of our next section.

Example. *Suppose that you have \$100,000 deposited with a local bank in the U.S., which is FDIC insured. The local bank has invested heavily in providing residential mortgages. A big economic downturn occurs, and many people lose their jobs and cannot make their mortgage payments. The bank gets in trouble and fails. In doing so, it declares its clients should expect to receive 40 cents on the dollar. How much money should you expect to receive back from your \$100,000 deposited?*

SOLUTION. *If your account is FDIC insured, then you are protected up to \$250,000; therefore, you should expect to receive back your entire \$100,000. However, if your account were not FDIC insured (not all accounts are treated the same, but savings accounts and checking accounts, the most typical examples, usually are), then you should expect to receive back*

$$100,000 \times 40\% = 40,000;$$

i.e., \$40,000.

4. Collateral

Collateral is a financial or physical asset that serves as a backup for the credit risk of the borrower. Suppose a borrower has a \$1,000 loan from a bank. From the bank's point of view, we learned that there's an uncertainty associated with recollecting this \$1,000. How can the bank be expected to bear this risk? One way they could do so is to require collateral for the loan. For example, suppose the loan went to purchase a car. The car serves as collateral. If the borrower continues to make payments in a timely fashion, then the borrower can continue to use the car. However, if the borrower fails to make timely payments, the ownership of the car effectively diverts back to the lender. This transfer is the purpose of collateral. The car collateralizes the loan. Defaulting on the loan means the bank gets the car instead of the money that was lent to buy it. This is an example of collateral that is physical.

Banks don't really want to own cars as they are in the financial industry and not the automotive industry. Banks do want to measure credit risk carefully so their customers can provide timely payments with minimal defaults. This allows them to focus on their core business, such as servicing depositary accounts and lending funds to qualified individuals and businesses. Banks are not the only financial institutions that can do this. What other institutions exist in the financial landscape? We will discuss this topic in the next section.

5. Financial Institutions

There are distinct types of financial institutions. One way to consider distinct kinds of financial institutions is to look at the types of activities in which they engage. In doing so, we can categorize financial institutions according to the way in which they provide their primary services:

1. **Depository Institutions.** Depository institutions include banks and their close relatives, such as savings and loan companies, credit unions, trust companies, and mortgage loan companies.
2. **Contractual institutions.** Contractual institutions are those whose businesses provide long-term contractual services. The main types of these are insurance companies and pension funds.
3. **Investment institutions.** Investment institutions are those financial institutions that either originate or manage ongoing investments. These include underwriters, investment banks, mutual funds, closed-end funds, hedge funds, and unit investment trusts.

The reason to know these distinct types of financial institutions is to understand that each offers different services. Keep in mind that some firms can easily be involved in all three services. For example, you may find a multinational bank that can:

- provide you with a loan or mortgage,
- maintain your property insurance program, and
- manage your investment account.

So, within multinational large financial institutions, it is likely that a multitude of services are provided. It is also easy to find companies that specialize in one of the services. For example, within the world of investment institutions, you may find a mutual fund that provides a way to own a collection of bonds that are professionally managed. The mutual fund will not be of any help, however, if you need a loan, a mortgage, or insurance. Furthermore, there are also new FinTech or disruptive types of businesses that are providing these financial services independently. For example, it is possible to find a FinTech company that allows you to transfer money from one place to another ("The 7 Best Transfer Money and Currency Exchange Apps"). Banks charge hefty wiring fees for this service, but FinTech companies, through apps on your smart phone, can do this for a fraction of the cost if not for free. Later in this module, we will distinguish between distinct types of financial institutions to better understand their features and benefits.

Certain large financial institutions are so important to the financial system that their demise could wreak havoc. For example, if a multinational bank were to fail, it could cause widespread fear and panic, leaving depositors without their funds, causing insurance claims to go unpaid, failing pension recipients who need their retirement income, etc. Can we trust these financial institutions to adopt best practices and ethical behavior? While an amount of self-regulation is helpful, it is even better to have regulation at regional, national, and even international and global levels. This regulation can provide stability and reassurance to promote a rehabilitative reliability and assurance that the financial institution acts in prudent and appropriate ways for the trust of the stakeholders in the system. Prior to the global recession, these institutions ran into trouble and looked for assistance to help bail them out of trouble. This term is known as **"too big to fail"**. As a result, indeed, some of them were too big to fail, meaning that their fall could have caused further collapse in the financial system. What helps to minimize or contain this risk? What would help to avoid the problems in the first place? Regulation.

6. Regulation

To understand regulation in the financial industry, first, let us simply think of regulation in a sporting industry. For example, consider football (or soccer as it is known in the United States). There are specific rules in the sport of football. These rules apply to games that are played by most nations in the world. These nations meet periodically on the football field to determine the winner of the World Cup. These rules are well known by players, enforced by referees, and often challenged by the coaches who find that they may have been broken to the disadvantage of their team. Imagine a football game without such rules: the fouls, the cheating, the kicks we don't see, etc. That would be a different game altogether. The rules and regulations are meant to ensure a safe, fair, and equitable game. They also increase reliability by ensuring that the winning team did so without resorting to cheating or using unethical and unscrupulous tactics. That builds confidence in the spectators and sponsors, knowing that the business model of advertising and marketing work in a trustworthy environment.

In the same way, financial institutions have regulation. Financial regulation is meant to ensure the safety and protection of the interests of stakeholders. These stakeholders include consumers, taxpayers, employers and employees, pension holders, investors, homeowners, small businesses, large businesses, and even the financial institutions themselves. This list comprises everyone in the financial system.

Financial regulation not only includes the rules and practices by which firms and individuals operate but also the enforcement of said rules. Just as in the game of football, the rules don't change, but there can be games where the referees selectively enforce rules, miss certain bad plays, or simply make bad calls. Likewise, in finance, some companies break regulatory rules, receive fines or penalties, and fail to comply with mandated policies. Regulation seeks to ensure that the financial system works in a smooth and trustworthy manner.

It is not always easy to determine whose authority a problem may fall under. It may be difficult to identify the culprit in markets with millions, and perhaps billions, of transactions. The complexity of trades, investments, disclosures, and informational asymmetries can create scenarios that have never occurred before. Our goal here is to simply introduce the concept of regulation. In the next topic, we will tackle how regulation can help to minimize credit risk.

7. Insured Accounts

When you think of finance, you can think of it as two-way interactions. Let's examine the *"assurance"* those accounts have from each side. Let us begin with the lender. The lender has lent funds to a borrower. How can the lender have assurance that the borrower will repay? As discussed, one of the contributions of regulation that came out of the global crisis is that there are more transparent qualifications needed to get the loan in the first place. The idea of a no-documentation loan is harder to achieve. Even though this might have been more "profitable" for the bank in the short run, the aggregation of multiple loans is a definite increase in the credit risk of the bank. Therefore, regulation can help assure the stakeholders that financial institution—its investors, its employees, and even other participants in the financial industry—that the financial institution is acting in a prudent manner in its lending.

Likewise, from the depositor's point of view, how can the depositor be assured that the bank is protecting its deposits? For this, we return to the depositary insurance corporation. There are many countries that have a type of deposit insurance system. A list of several of these countries with the max amount insured can be found at the following [link](#) from Wikipedia. These countries provide not merely *assurance* but *insurance* that if anything happens to a deposit (up to a specified monetary amount), then the depositary insurance corporation guarantees the amount. Therefore, borrowers can sleep well at night knowing that their deposits are safe. For example, in South Korea, each depositor has protection of up to 20 million Korean won. For each insurance policy holder, there is protection of up to 50 million Korean won. The idea of this insurance coverage helps to provide stability and trust in the financial system. Therefore, the idea of putting your money into a savings account or a certificate of deposit or other insured account is a very safe investment.

But what if you are interested in more than just saving money and earning interest? Is there a way that you could also invest in the fixed-income market? This topic will be addressed in the next lesson: bonds.

8. Basics of Bonds

What is a bond? A bond is a debt security by which the issuer/borrower is committed to paying back to the bondholder/lender the cash amount borrowed (the **principal** or **bond notional**) in addition to periodic interest calculated on the amount of a principal during a given period of time.

Let's see how a bond works with an example.

EXAMPLE *The U.S. Treasury has issued a bond today (1/1/2024) with the following key characteristics:*

- maturity date 12/31/2025;
- coupon rate 4%;
- nominal issued amount \$20,000,000,000; and
- interest payment is semi-annual (typical of U.S. bonds, unlike Euro Treasury bonds that usually pay interest once a year).

Write down the interest payments and the principal repayment with the relative dates.*

SOLUTION

- interest payments:

$$20,000,000,000 \times \frac{4\%}{2} = 400,000,000;$$

i.e., the U.S. Treasury will pay \$400,000,000 every six months, or on 6/30/2024, 12/31/2024, 6/30/2025, and 12/31/2025.

- principal repayment: \$20,000,000,000 on 12/31/2025.

Note that to compute the interest payments for bonds paying semi-annual coupons, we take the coupon rate and divide it by 2.

Why would the U.S. government, via the Treasury Department, or a corporation issue bonds? Well, in the case of the U.S. government, it is a way to finance projects of public interest, like the construction of roads, bridges, and schools. In the case of corporations, it is a way to finance investment projects at a cost (interest rate) that is lower than the return on those investment projects.

We can also look at bonds by saying that a borrower (debtor) has issued an I.O.U. to the bond buyer (creditor). In other words, the borrower has received a stated amount of money up front, the *face value* (also known as notional value) of the bond. In exchange for this principal, the bond issuer commits to providing timely interest payments, known as *coupon* payments, and returning the principal at maturity to the bond buyer.

So, if we wanted to receive some money, we could do one of two things.

1. We could invest our money in a certificate of deposit. In this investment, we are giving money to a bank, which holds it for a specified period of time and then pays us interest and the principal back at the term of maturity.
2. We could purchase a bond. For this, we go to a specific issuer from whom we buy the bond. We might pay them par, or the face value, for a 2-year bond. Then, over the next two years, we would receive coupon payments periodically and the principal back at the end of two years. These amounts are fixed, assuming no defaults.

So, what are the differences between our savings example and the bond example? Both opportunities are examples of **fixed income** investments. It is called fixed income for the two reasons that comprise the term: that the rates being provided are **fixed** and that there is a stream of **income** consisting of the **interest** payments. The repayment of principal is not so much income as it is just the return of your original investment. In other words, so long as the issuer does not default, the bond will repay the amounts just as the bank would repay the amounts.

The difference between the two is that the bond is an asset and not merely an account. The bond can be traded. It is something for which you can see a market price. In the savings example, the CD is not a traded asset. Its value does not fluctuate over its lifetime. You cannot cash it out without severe penalties. The bond, on the other hand, can be sold to someone else, so in a sense, you can get its value liquidated to cash before the term to maturity. That is, the bond does not need to be held to maturity.

To review, there are three ways to get paid from holding a bond:

1. Wait until a coupon date to receive the coupon payment.
2. Wait until maturity at which you receive the principal repayment.
3. Sell the bond before the term to maturity at the prevailing market price.

Since it is an asset, it can be sold in the marketplace. And so, because the bond can trade, its price can fluctuate with the market. What is it that drives the price of bonds? This is the topic of our next lesson.

9. Conclusion

In this lesson, we examined the process of default and the broader picture of credit risk. We looked at how collateral, regulation, and insurance all help to keep this credit risk in check. We ended by introducing bonds. In the next lesson, we'll expand our coverage on bonds.

References

- "The 7 Best Transfer Money and Currency Exchange Apps. Future of Fintech," *Wingjie*, 9 Feb. 2021, [wingjie.com/2020/10/26/the-7-best-transfer-money-and-currency-exchange-apps-future-of-fintech](#).

BUYING AND SELLING SHORT

Reading Time	150 minutes
Prior Knowledge	Interest, Interest Rates, Term To Maturity, Bond Definition
Keywords	Discounting, Credit Risk, Buy-Side, Sell-Side, Shorting, Financing

In the previous lesson, we examined ways in which the lender can have their risk mitigated. We also saw that aside from loans, a borrower can issue a bond, which would then make interest payments to the bond holder. In this lesson, we examine bonds in more detail. We'll discuss buying a bond and selling a bond short. Note that this is different from selling a bond. We'll conclude by studying the bid-ask spread.

1. Bond Pricing via Discounting Cash Flows

What is the price of a bond? A bond entitles us to receive a certain number of cash flows at pre-specified times. In our earlier example at the end of Lesson 2, for instance, the 2-year Treasury bond made 4 interest rate payments and 1 principal repayment at specific dates. One could be tempted to say "let's sum up the value of all the individual cash flows and that is the value or price of our bond". Well, not quite because the cash flows happen at different moments in time and we are by now aware that there is such a thing as the time value of money. So how do we treat different cash flows at various times? We already have the mathematical tools to solve this problem, and we will do so in the next section of this lesson.

What are we going to do to find the present value of all the relevant cash flows at different times. Doing this requires that there is a discount rate at which we can convert future cash flows to their value (of money) today.

Suppose we will receive \$1,000 in one year and that the interest rate at which we discount is 4%. Then what is \$1,000 in one year worth in today's money? The formula for present value tells us that the present value of \$1,000 in one year is

$$1,000 \times \frac{1}{1 + 4\%} = \textbf{\$961.54}$$

The number $\frac{1}{1+4\%} = \textbf{0.96154}$ is the discount factor to use for 1 year if we assume that the relevant interest rate to use is 4%.

Similarly, if we are expected to receive \$3,000 in 2 years, what is the value of \$3,000 in 2 years in today's money? Assuming that the relevant interest rate is always 4%, then at the discounted value today, \$3,000 in 2 years would be

$$3,000 \times \frac{1}{(1 + 4\%)^2} = \textbf{2,773.67}.$$

The number $\frac{1}{(1+4\%)^2} = \textbf{0.92456}$ is the discount factor to use for a 2-year investment if we assume, again, that the relevant interest rate to use is 4%.

We are now ready to price a bond.

2. Actual Pricing of a Bond

We show how to actually price a bond using an example and detailing all cash flows and calculations.

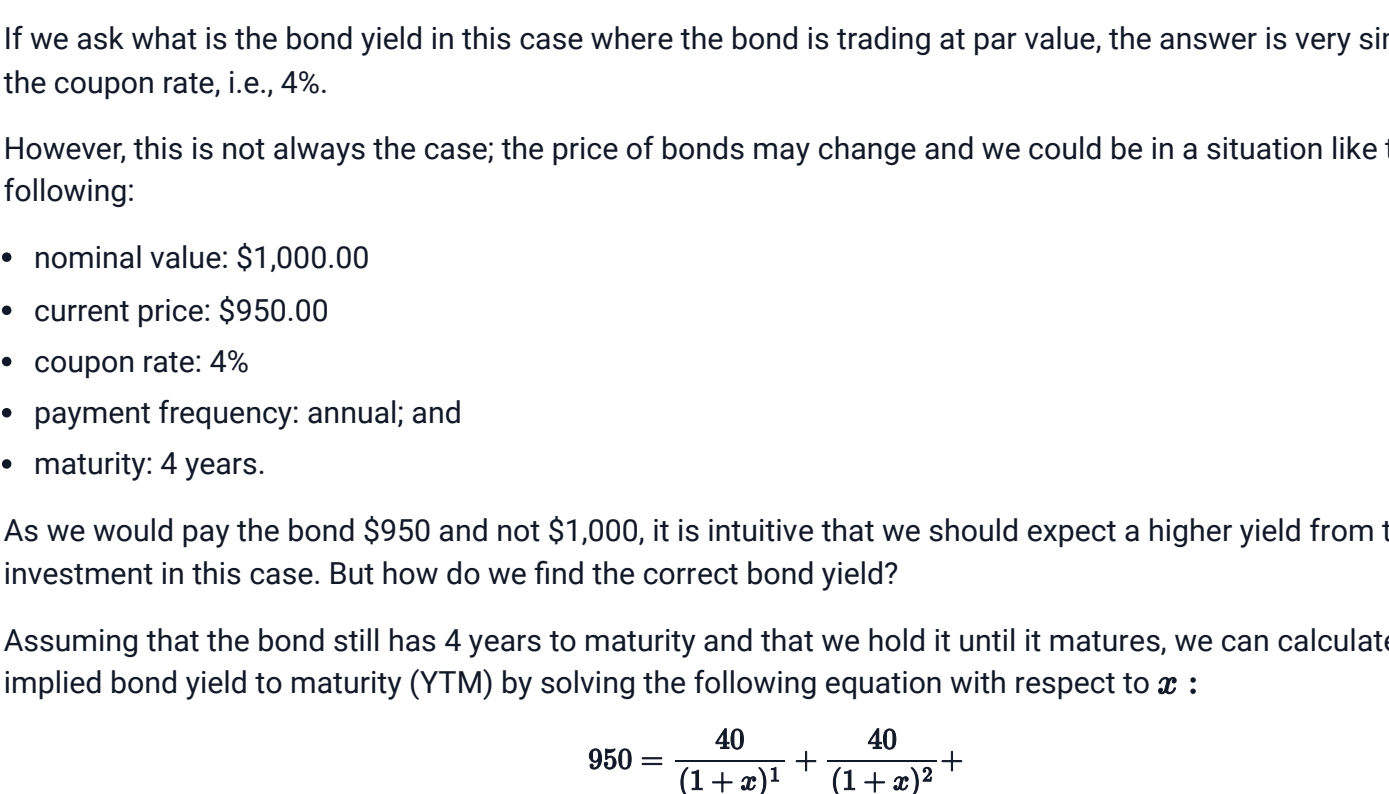
Suppose that Bond PQU has the following characteristics:

- nominal amount (par value): \$1,000,000;
- coupon rate: 4.5%;
- coupon payment frequency: annual
- maturity: 5 years from now
- current level of interest rate 5% (this will be our relevant interest rate for discounting).

We start by listing all the relevant cash flows:

- in 1 year, coupon interest payment of \$1,000,000 X 4.5% = \$45,000;
- in 2 years, coupon interest payment of \$45,000;
- in 3 years, coupon interest payment of \$45,000;
- in 4 years, coupon interest payment of \$45,000;
- in 5 years, coupon interest payment of \$45,000 plus the principal repayment of \$1,000,000.

Figure 1: Cashflows Add Up Every Year.



Now, that we have all the cash flows well organized, we need to discount each of them and then we sum that up. As such, the value of the bonds can be calculated as:

$$\begin{aligned} & 45,000 \times \frac{1}{(1 + 5\%)^1} + 45,000 \times \frac{1}{(1 + 5\%)^2} + \\ & 45,000 \times \frac{1}{(1 + 5\%)^3} + 45,000 \times \frac{1}{(1 + 5\%)^4} + \\ & (45,000 + 1,000,000) \times \frac{1}{(1 + 5\%)^5} = \textbf{978,352.62} \end{aligned}$$

Hence, we found out that the price of this bond

(at time 0, today) is \$978,352.62, less than the nominal value of \$1,000,000 at which we bought it. When this happens, we say that the bond is trading **below par**. If the bond traded above the par value, we would say that the bond is trading **above par** instead.

You can change discount rate, coupon rate, and even principal amount in the following [spreadsheet](#) and find the price of the bond.

Why is the price of the bond trading below par in our example? Well, our bond pays a coupon rate of 4.5%, but the current coupon rate level is 5%. Now, would you invest \$1,000,000 to pay a bond with a coupon rate of 4.5% when you could buy a different one that pays a coupon rate of 5%? I bet that your answer is no. In other words, if you want to sell the bond to someone instead of holding on to it until maturity right now, you have to sell it below par or nobody would buy it.

An attentive student has probably noticed that the mathematical reason for the price being below par is that we are discounting interest payments at a higher rate than the coupon rate.

Go to the spreadsheet above and try changing 5% to 4%, 6%, or 4.5% and notice what happens to the bond price. Essentially, higher interest rates mean lower bond prices.

3. Implied Yields to Maturity

The **coupon rate** or **bond yield** for a bond trading at par (i.e. at face value) is very easy to calculate:

$$\text{coupon rate} = \frac{\text{annual coupon payment}}{\text{bond face value}}.$$

Suppose that we have a bond with the following characteristics:

- nominal value: \$1,000,000;
- current market price: \$1,000,000;
- coupon rate: = 4%;
- interest payment frequency: annual; and
- maturity: 4 years.

If we ask what is the bond yield in this case where the bond is trading at par value, the answer is very simple as it is the coupon rate, i.e., 4%.

However, this is not always the case; the price of bonds may change and we could be in a situation like the following:

- nominal value: \$1,000,000
- current price: \$950.00
- coupon rate: 4%
- payment frequency: annual; and
- maturity: 4 years.

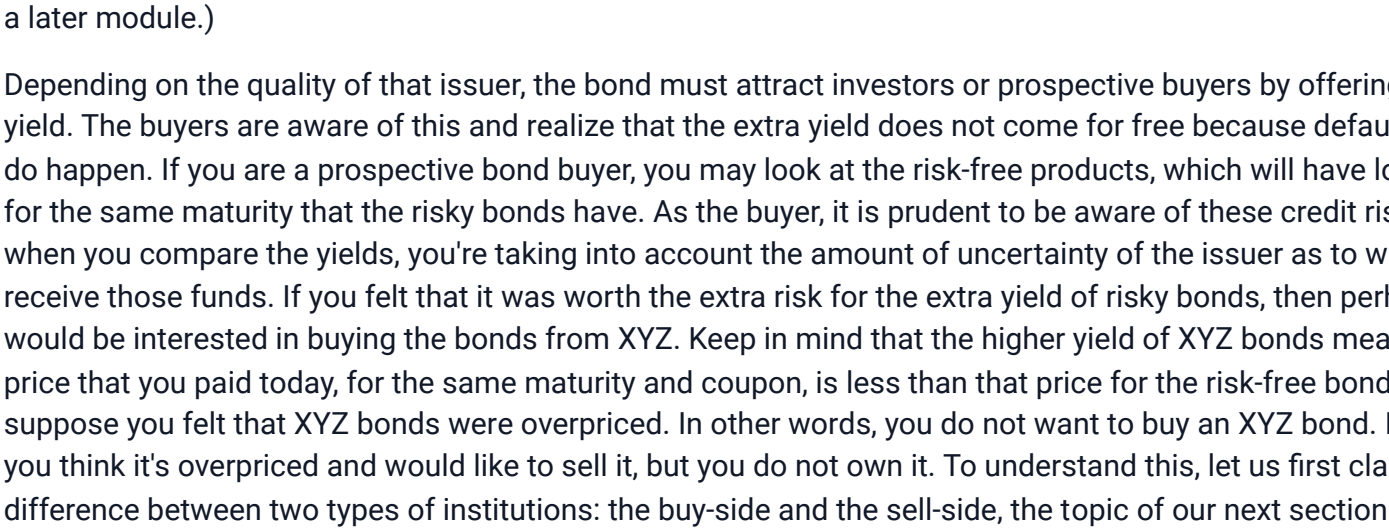
As we would pay the bond \$950 and not \$1,000, it is intuitive that we should expect a higher yield from the investment in this case. But how do we find the correct bond yield?

Assuming that the bond still has 4 years to maturity and that we hold it until it matures, we can calculate the implied bond yield to maturity (YTM) by solving the following equation with respect to x :

$$\begin{aligned} 950 = & \frac{40}{(1+x)^1} + \frac{40}{(1+x)^2} + \\ & \frac{40}{(1+x)^3} + \frac{1040}{(1+x)^4}. \end{aligned}$$

This is a highly non-linear equation, but it can be solved using numerical analysis or a spreadsheet (for instance Google Seek in Google Sheets). Doing this shows that the implied yield to maturity is 5.424%, quite a bit more than the coupon rate of 4%. You can use this [spreadsheet](#) to solve the problem.

Figure 2: Inverse Relationship between Bond Price and Yields



Is this all the math we need for bonds? We are close but not quite there yet. So far, we have assumed that the bonds pay annual interest rate coupons. What if they paid those coupons semi-annually as is mostly the case in the U.S?

The principles behind pricing a bond do not change. However, the formula is a tad more complex, and we will provide it in a moment.

Let

- P be the price of a bond that we want to compute;
- n be the number of years to maturity;
- c be the annual coupon rate;
- α be the current annual prevailing rate;
- M be the principal amount (also known as the notional or par amount) of the bond; and
- the interest payment frequency be semi-annual.

Then, the price P of the bond is found using the following formula

$$P = \sum_{j=1}^{2n} \frac{M \times c/2}{(1 + \alpha/2)^j} + \frac{M}{(1 + \alpha/2)^{2n}}.$$

Note that in the formula, all interest rates are divided by 2 and we have $2n$ payments as the bond pays two coupons every year.

The last equation can be easily used in a spreadsheet if one does not feel inclined to do some geometric series math.

Also, if we were given the current market price of the bond, say P_0 we could solve the equation above with respect to x and determine the implied YTM associated with a price of P_0 for the bond.

Now we know enough bond mathematics to solve many bond problems.

A serious student may be asking whether it is typical that the coupon rates and the current interest rates are always fixed. In many cases the coupon rates are fixed, but there are bonds for which the coupon rates could be indexed to account for inflation (i.e., like paying a real rate instead of a nominal one). Also the current rates, which we use for discounting, can and do change over time. This is not the end of the world. Yes, we will no longer be able to get nice elegant formulae to do our calculations, but spreadsheets are still viable tools. We can indicate the general formula, which would become

$$P = \sum_{i=1}^T \frac{CF_i}{(1 + r_i(0, t_i))^{t_i}}$$

and which allows for the cash flows to change with the index i and for the interest rate we use to discount the cash flows to change with the length of time.

How do we get the $r(0, t_i)$ that appear in the formula? They are usually found from **zero-coupon bonds** of different maturities, if they are available and "zero".

A zero-coupon bond, also known as a "treas", is a bond that does not make interest payments. Instead, it trades at a (usually deep) discount to its face value. For instance, we could have a zero-coupon bond like the following:

- face value: \$1,000;
- maturity: 10 years; and
- current market price \$630.

In other words, we can buy it for \$630, we will receive no interest payments (note that a coupon rate is not specified), and in 10 years' time, \$1,000 is returned to us. We can calculate our yield very simply using the math of present and forward values seen in Lesson 1:

$$\text{yield} = \left(\frac{1000}{630} \right)^{1/10} - 1 = \textbf{4.73\%}.$$

Assuming that this zero-coupon bond is issued by a reliable issuer and it is fairly liquid (so that its market price is also reliable), we could use the result just found to say that $r(0, 10) = \textbf{4.73\%}$.

It is worth mentioning that, while zeroes carry the usual market risks for investors, in particular when dealing with long maturities, they also contain a possibly sizable credit risk as the company that issues the bond could end up defaulting and without having paid investors any interest.

When we look at a bond issued by the UK Treasury or the U.S. Treasury, we see that the yield for a two-year bond is 3%. However, if we were to look at a bond issued by a company, call it XYZ, then we see that the yield is 4%. Why would two different issuers have two different yields for the same maturity? The answer is an old friend: credit risk, the topic of our next section.

4. Credit Risk

Credit risk is always an issue for a lender. Consider the following situation: When you are depositing funds in an account, you are "lending" the bank your funds for a period. Remember, you are protected by institutions such as the depositary insurance corporation. When you are a bond buyer, you are also a lender. The bond buyer is supplying the funds up front to the bond issuer. The bond issuer is the one who is borrowing. They are the ones who are making prompt interest and principal payments. Therefore, the credit risk is born by the bond buyer, which is acting as the lender here.

From the bond buyer's perspective, why would they want more yield from the corporation than they would from the federal government? The answer comes, in part, from credit risk. You see, the credit risk of the federal government bonds is effectively zero. That is because the federal government has protections that will eliminate the chance of default. (There are historical cases where countries have defaulted on their bonds, but at this time, we'll assume government bonds are risk-free). This is why the yield associated with government-issued bonds is called the risk-free rate. So, when you look at bonds issued by other governments outside your country, these are also called risk-free rates in their sovereignty. The risk-free rate return refers to an interest rate for a given maturity that is default free. That means that the issuer is considered to have zero probability of defaulting on the payments.

Certain federal governments can safely be considered risk free because they have advantages that corporations do not have. These advantages include setting monetary policies, printing new currencies, and increasing taxes. (Note that not all countries have debt that is considered to be risk free). These revenue-generating actions can ensure that they have sufficient funds to meet their liabilities. However, the company XYZ does not have these activities available to them. They could either increase revenue, cut costs, or issue new longer-term debt to pay off short-term debt. Therefore, XYZ-issued bonds have some **probability of default**. (We'll talk much more about this in a later module.)

Depending on the quality of that issuer, the bond must attract investors or prospective buyers by offering a higher yield. The buyers are aware of this and realize that the extra yield does not come for free because defaults can and do happen. If you are a prospective bond buyer, you may look at the risk-free products, which will have lower yields for the same maturity that the risky bonds have. As the buyer, it is prudent to be aware of these credit risks so that when you compare the yields, you're taking into account the amount of uncertainty of the issuer as to whether you receive those funds. If you felt that it was worth the extra risk for the extra yield of risky bonds, then perhaps you would be interested in buying the bonds from XYZ. Keep in mind that the higher yield of XYZ bonds means that the price that you paid today, for the same maturity and interest, is less than that price for the risk-free bond. But suppose you felt that XYZ bonds were overpriced. In other words, you do not want to buy an XYZ bond. Instead, you think it's overpriced and would like to sell it, but you do not own it. To understand this, let us first clarify the difference between two types of institutions: the buy-side and the sell-side, the topic of our next section.

5. Buy-Side vs. Sell-Side

The terms buy-side and sell-side at first seem to convey the idea of one side buying assets and the other side selling assets. This, however, is a misunderstanding. The buy-side refers to the market that is investing on behalf of others. The buy-side includes institutions that collect funds from investors, workers, pension holders, labor unions, etc., and seeks to invest on their behalf, with their stated investment interests and risk tolerances. The buy-side, in other words, is professionally managing client money. At times, those investment managers may decide to buy investments. At times, they want to realize the capital gains on those investments and sell the investments. As such, the buy-side is both buying and selling securities, but they are doing so on behalf of investors.

The sell-side, on the other hand, is NOT acting on behalf of investors but rather is acting on behalf of the investment bank. The sell-side's primary responsibility is to make markets. **Market-making** means that they are willing to engage with the buy-side (or even other sell-side firms) by both offering securities to sell and bidding on securities to buy. You see, the buy-side is trading on behalf of customers; they are looking for opportunities that appeal to their customer base. The sell-side, on the other hand, is simply trading according to the needs and desires of the buy-side and other sell-side firms who want to trade. Sell-side trades make what is known as a **two-sided market**.

How does each side effectively make money? In broad terms, the buy-side makes profit by earning a fee for managing the funds. In addition to a management fee, they may collect fees related to the trading in the account or consulting with clients, and they might collect a performance incentive fee. If the fund earns above a certain amount (which may be 0%), then the fund may take a percentage of the amount earned. Effectively, the buy-side acts as a fiduciary for its clients by offering professional management and is paid both a fixed fee and potentially an incentive fee for good performance.

The sell-side, however, makes money in a more complicated way. To understand this, let us look at an example by considering a buy-side firm called PQD. Let us say that they wish to set up a brokerage account with a sell-side firm called ABC. ABC is willing to do two things at any given time:

ABC is willing to sell shares of our risky bond XYZ at \$101.00

ABC is willing to buy shares of our risky bond XYZ at \$100.50.

(Of course, that price may be displayed in a yield term.)

The price at which ABC is offering to sell is known as the **offer price**. Like everything, there is also another way to refer to this: the **ask price**. We will use offer price and ask price interchangeably since they mean the same thing.

The price at which ABC is willing to buy is known as the **bid price**. In the next section, we will discuss the difference between these.

6. Bid-Ask Spread

The bid price is the price at which someone is willing to buy. The ask price is the price at which someone is willing to sell. The difference between these two prices is known as the **bid-ask spread**. The sell-side makes a profit by this bid-ask spread. Imagine that there was no bid-ask spread: you would be able to buy and sell at the exact same price. So which financial institution could afford to supply trading to market participants without earning any money for it? Well, none.

If you have ever been to an airport kiosk that provides a foreign exchange service, then you will understand this. Suppose you arrive at the airport wanting to convert euros to Japanese yen. Your 100 euros bought you 12,800 yen. Then, you find out that your flight got cancelled. So you return to the airport kiosk and convert the 12,800 yen back to euros, but you only get 915 yen. The kiosk makes money by selling you yen at a high price and buying back the yen at a low price. If you were to do this repeatedly, you would end up out of money. The difference between the two prices is known as the bid-ask spread. This is how the foreign exchange kiosk attendant earns money. In addition to the bid-ask spread, they may have market transaction costs, such as commissions or transaction fees.

The same ideas apply to sell-side market makers. The bid-ask spread, and any transaction costs, are part of the revenue of the sell-side. You can see that the sell-side has a tough job because to buy and sell, they must have inventory amounts that may drive them to either hold too much of a security or run out of it and borrow the security. Indeed, it is possible to sell things that you don't even have. So, how is it possible to sell securities that you do not have? How is it possible that the sell-side can consistently sell a security, even if they do not have any? For this, we turn to the next section: shorting.

7. Shorting

One of the lessons you learn early in finance is that you should "buy low and sell high." This means that if you see a security that you like, you use funds that you have to purchase it, hold onto it until the price goes up, and then sell it at a higher price, realizing the capital gain as a profit, less any fees like bid-ask spreads and transaction fees.

The idea of buying low and selling high is very straightforward, but it is less so in that the order can be reversed. In other words, what you want to "sell high, then buy low?" This is indeed a trading strategy. But however can you sell something if you don't own? This is known as **shorting**. **Shorting** refers to borrowing a security that you do not own, selling it in the marketplace, and receiving cash for it. The **short seller** then hopes and waits for the price to drop; if and when it does, the short seller buys the security at this lower price. When that happens, the short seller is **covering the short**. Once they take possession of the security, they return it to the party that lent it to them. If they sold at a high price and covered at a lower price, they still make a profit: selling price - buying price.

So shorting is somewhat of a complex transaction. It involves:

- Borrowing something that you do not own.
- Selling it in the marketplace.
- Paying the lender (owner of record) of the security a fee (that increases with time).
- Buying (covering) the short later.
- Returning the security to its original owner.

Short sellers trade because they think a security is overvalued. They prefer to sell it now at what they believe is an inflated price. They receive cash for it, which they can deposit in an account for any costs associated with the short. If all goes well, then the price does indeed drop and the short seller can cover the security, return it to the owner, and complete their responsibilities. Doing so has earned them whatever the capital gain is (high – low), less any transaction fees.

The lender of the security, in some ways, is like the lender of our loan examples. The borrower of the security, the short seller, is like the borrower of our loan examples. Therefore, we have seen this familiar scenario before. Instead of lending cash, we are lending securities. What is the same and what is different?

In principle, it might seem that borrowing shares creates a credit risk for the lender in the sense that the borrower may not be able to return the securities (like the cash borrower who is not able to pay back the loan). However, selling short is a bit more complicated and the credit risk is quite small if the shares are borrowed. To be able to borrow shares, we will need a **margin account**, i.e., an account with a brokerage firm that allows us to borrow money in order to buy stocks. **Borrowing shares** is, for all practical purposes, like a margin loan on which we may pay interest on an outstanding debt (the value of the shares borrowed). To be able to open a margin account, we would need to be approved by the brokerage firm.

Furthermore, in the U.S., to be able to make trades in our margin account, we need to have cash or stock equities in the same margin account as collateral in an amount at least equal to 50% of the short position. Once we receive the shares and sell them, we won't be able to liquidate the cash we receive from the short sale. If at this not enough, we will be required to keep enough equity in the account to serve as collateral, which is at least 25% per exchange rules, but our broker may set the margin higher, depending on how risky the stock is and the value of our short position. Hence, if the value of the stock goes up instead of down, we may be asked to inject further collateral on our margin account, or the broker can liquidate our position and return the shares to the original owner.

Example.

- Shares of MNQ are currently trading at \$80/share.
- Fund ABC's analysts believe that the shares are overvalued and that they should fall to \$60/share.
- ABC's portfolio manager decides to short the shares.
- The portfolio manager contacts the fund's broker who is able to locate (that is the technical term used for finding the many shares) of MNQ. The broker has 500,000 shares to lend at an annual interest rate of 5% as there are not many shares of MNQ available.
- The 500,000 shares are sold short at \$79.20/share.
- After 6 months, the shares of MNQ have fallen but not as much as hoped, and they are trading at \$72/share. The portfolio manager decides to unwind the position and buy back the shares, which are purchased back at \$72.33/share.

Compute Fund ABC's profit from this investment, assuming no interest payments to ABC's cash in margin account.

Answer. The amount invested (borrowing the shares) is

$$500,000 \times \$80 = \textbf{\$40,000,000.00}$$

The proceeds from the short sale are

$$500,000 \times \$79.20 = \textbf{\$39,600,000.00}$$

When we buy back the shares after 6 months, we spend

$$500,000 \times \$72.33 = \textbf{\$36,165,000.00}.$$

Hence, the profit from the short sale is

$$39,600,000.00 - 36,165,000.00 = \textbf{\$3,435,000.00}.$$

The amount of interest we need to pay for having borrowed the shares for 6 months (financing cost) is (using compound interest)

$$40,000,000 \times (1 + 5\%)^{0.5} - 40,000,000 = \textbf{\$987,803.06}.$$

Hence, the profit from the trade is \$2,447,196.94.

8. Financing Short Positions: More Details

Someone is really willing to lend you a security if there is something in it for them. Whoever lends out the security will want to receive an amount of income for it. This is known as the **financing cost**. Imagine that you are the owner of XYZ Bond. As a bondholder, you make money by receiving interest payments. However, there is another way you can make money. You can agree that your broker on the sell-side can lend out your bonds to another company. When you do so, you will receive whatever the financing cost is from that short seller. Financing provides extra income to the owner of a security.

When someone lends out the security, they want to be assured that they receive this financing rate from the short seller. In addition, they also get the interest from the bond as if they still owned it. In other words, the short seller sells the bond to another customer. The person that buys those bonds now owns them. The original owner merely has an "I.O.U." from the short seller. However, the true cost of shorting requires you to make good on all income that the ownership bestowed. When the issuer makes a coupon payment, the short seller is expected to make that payment, as part of the financing cost, to the original owner. Otherwise, the owner would have forfeited that interest income. Remember, the original bond holder is looking to make extra income by lending out their bonds, not substituting interest income for income from financing. So, the cost of financing is two-fold. If you are shorting a bond, then you have two costs to deal with:

- The financing cost, which is effectively an interest rate for borrowing the security.
- The income cost of whatever that security produced. Since you borrow the bond, you are responsible for making that interest payment to the original owner!

To emphasize: the person who lent you the bond still is eligible to receive the interest, but they do not have current ownership of the bond. Instead, they have an "I.O.U." the short seller's promise to make good on the interest payments. As such, when you are shorting an investment, you must cover the lost income generated as well as a financing cost for the privilege of borrowing the bond.

For short sellers, it is very important to find bonds for which they believe the price will drop far enough to offset both the financing costs and income cost, as well as other transaction fees, and to ensure that a profit can still be earned. Doing so gives truth to the statement "sell high and buy low."

Shorting is a remarkably interesting part of the market because it serves the purpose of preventing assets from getting overvalued. If there's consensus that a financial asset is overvalued, then short sellers come into the market and put downward pressure on that asset, helping it to achieve its natural equilibrium state. If there is no mechanism to short, then a market can become frozen. That means that sellers have prices that are too high, and there are no buyers willing to pay the prices.

Does this sound unrealistic? Think of commercial real estate. There are many properties throughout the world that remain vacant because property owners expect to receive an amount for them that exceeds what businesses are willing to pay for them. As a result, there is not a good consensus of what the price is. To put it in terms that we just learned, the offer price for commercial real estate often exceeds the bid price by such a large amount that properties stagnate. This means that property owners are looking for an offer price that is much greater than the prospective tenants are willing to pay. The offer price exceeds the bid price. There is no middle ground; they do not meet halfway. They simply do not come to terms. Consequently, the real estate remains vacant, often for years.

Shorting is a risky business because of the potential for unlimited loss. When you think about the fact that prices can jump up, there are potentially catastrophic losses to the short seller. We have been discussing bonds, but imagine you apply this to other asset classes for which there are no natural ceilings to the prices, like stocks or Bitcoin. During the global recession, there was a lack of confidence in the financial institutions themselves. It got to a point where so many investors were shorting the financial institutions that the government intervened and prohibited short selling of financials. This helped to restore some faith and trust in the financials and was part of a more comprehensive solution to avoiding a global depression. This was one of the actions that regulatory agencies suggested. In the next lesson, we will talk about these things first by focusing on central banks.

9. Short Squeezes

At this point we should be aware of the reasons behind short selling. Unlike what many non professional people believe, short selling is not just an attempt at manipulating the market. In fact, when speculators may sell short with the hope of realizing a profit on, say a stock that appears overvalued, this is not unlike what long investors do when they bet on the price of an asset to go up. The difference is that short sellers expect the stock to go down in price while long investors expect the stock to go up in price, instead. Furthermore, there are also traders and fund managers who use short selling to offset certain risks in a long position or a portfolio. Finally, short sellers contribute to creating liquidity for trading.

However, short selling is not without risks, quite the opposite. The most feared of these risks is arguably a **short squeeze**.

To understand a short squeeze well, one has to be aware of at least the following three facts.

- The **short interest** represents the number of shares that have been sold short and are still outstanding (i.e., not yet covered). When a trader borrows shares of a security from a broker and then sells those shares to some buyers, these shares are counted as short interest.
- The number of shares that have been sold short is often expressed as a percentage on the public float and is an important element as it can signal market sentiment. In particular, an increase in the short interest may signal a bearish sentiment. A decrease, on the other hand, may tell investors that the market is overall becoming more bullish on a specific asset.
- It is possible for short interest to be more than the entire public float of shares of a company. Some may wonder how this is possible. Well, we

Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation should take you directly to the reading or to a page where you can download it.

1. Fotia, Alessio, and Ulrich Bindseil. *Introduction to Central Banking*. Springer, 2021.

<https://link.springer.com/book/10.1007/978-3-030-70884-9>

- **Required pages:** Chapter 1 (pp. 1-9 and 25–31)
- **Estimated reading time:** 1 hour

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

SURVEYING THE FINANCIAL INDUSTRY

Reading Time	60 minutes
Prior Knowledge	Bonds, Interest Rates, Credit Risk
Keywords	Central Bank, Monetary Policy, Investment Banks, Market Risk, Reinvestment Risk, Default Risk

In the previous module, we looked at bonds as another security in which lenders provide funding to borrowers. We also saw that buy-side and sell-side institutions comprise the financial industry. In this lesson, we examine different types of banks. We'll also look at different types of risks associated with bonds.

1. Central Banks

While we have discussed distinct kinds of banks, there is one type of bank we have yet to discuss: the central bank. Central banks are financial institutions whose purpose is to maintain stability and transparency for countries' economies. Central banks help to keep inflation in check, provide maximum employment opportunities, and enforce healthy monetary policy for a stable economy. Please refer to the [required reading by Fotia and Bindseil, Chapter 1 and Chapter 3](#).

The responsibilities of central banks are complex and tend to vary by country. However, one responsibility of central banks is to help set operational targets. As you review the required reading, note that the ultimate targets are the inflation rate and the foreign exchange rate. Central banks can set interest rates at which banks borrow and lend from each other. By doing so, they can help ensure that there are sufficient funds that can be provided between banks for them to conduct their business.

2. Investment Banks: Functions, Maturities, and Products

Different from central banks are investment banks. Investment banks have distinct functions:

1. They underwrite securities.
2. They help with IPO issuance.
3. They engage in sales and trading with investors.
4. They provide research for clients.
5. They bundle and sell securitized products.
6. They perform mergers and acquisitions.

Investment banks will also help **underwrite new businesses**. One thing that they must do is provide credit to businesses. Once again, in our module, we see credit risk. Investment banks continually monitor the credit risks of their clients, of their investments, and of each other. If you are a business, then you have a measure that will constitute your credit score. Investment banks will look at this credit score to determine the amounts and even rates at which you are charged for the use of funds.

Another purpose of banks is to **provide credit to individuals**. Individuals need a quantitative measurement of their credit. For example, in the United States, there is something known as a FICO score. FICO is an acronym for Fair Isaac's Company. This company developed a quantitative model that considers multiple factors in determining the credit worthiness of an individual and distills them down to a single number. That number is on a spectrum ranging from 300 to 850. Based on someone's FICO score, a person has credit worthiness that is made readily available to lenders: credit card lenders, mortgage lenders, property owners, and other financial institutions. Imagine the challenging work required if each prospective stakeholder in those examples would have to conduct their own credit risk assessment of an individual. These credit scores help recognize and reward individuals who build and maintain good credit by allowing them the use of credit at rates that reflect their creditworthiness.

Banks can also provide an array of fixed-income products ranging from very short-term loans and certificates of deposit to intermediate and advanced products:

- Short-term: typically less than 1 year.
- Intermediate-term: typically 1 to 10 years.
- Long-term: typically more than 10 years.

There are distinct types of products, including:

- Commercial paper
- Certificates of deposits
- Treasury bills
- Treasury notes
- Treasury bonds

3. Commercial Banks

Most of us are perhaps more familiar with commercial banks. We tend to think of them as places to deposit money, get a credit card, and get loans and/or mortgages. Commercial banks were highly regulated and their daily activities were integrated inside a wider financial system. Specifically, commercial banks were mostly concerned with attracting deposits from the public and lending money to individuals and businesses. In the U.S., the *Glass-Steagall Act* of 1933 made sure that there was a clear distinction between commercial banks and investment banks. That had important consequences for big banks like JP Morgan, which had to cut their services and lost several sources of income. Banks tried to sidestep the Glass-Steagall Act by using holding companies to avoid the restrictions on bank branching or having multiple offices. The *Bank Holding Company Act* of 1956 gave the Federal Reserve more oversight of this. In 1999, the Glass-Steagall Act was repealed, and today, the distinction between investment banks and commercial banks, for practical purposes, has little or no meaning—so much so that currently JPMorgan Chase Bank is among the largest commercial banks in the U.S. and also among the leaders in underwriting IPOs.

Many believe that the repeal of the Glass-Steagall Act allowed a consolidation (via mergers and acquisitions) of investment and commercial banks. Such consolidation in turn led to financial institutions that heavily invested in mortgage-backed securities, which, as we know, led to the big subprime mortgage crisis and its ultimate implosion in 2008-9. In fact, before the repeal of the Glass-Steagall Act, investment banks (the banks that generated the subprime mortgage derivatives) would not have been able to find themselves involved in mortgage lending as well (typically done by commercial banks). However, there are also those who counter that this is not true because big players like Lehman Brothers, Bear Stearns, and Goldman Sachs were big players in the meltdown of subprime mortgage investments, yet they were never commercial banks.

It has been theorized that the subprime mortgage meltdown was actually caused by the Department of Housing and Urban Development, which required Fannie Mae and Freddie Mac to purchase more affordable mortgages to encourage lenders to make loans to low-income and minority borrowers. Commercial banks, to meet HUD's objectives, began to give out mortgage loans to people who had no money for down payments or accepted unemployment benefits as a valid source of income. However, many of these lenders were not banks. Countrywide Financial, one of the main culprits, was not a commercial bank strictly speaking as it did not accept deposits. So, again, it is hard to argue that the cause of the subprime mortgage crisis should be attributed to the lack of distinction between commercial and investment banking.

Countrywide Financial is an ideal example to use to introduce some of the arguments for our next section: the existence of risks. Countrywide, under CEO Angelo Mozilo, pumped out complex mortgages to borrowers who did not understand them and could not afford them. Borrowers simply did not understand the risks involved with their mortgages, and Countrywide did not care much about what happened after the borrowers signed their mortgages because it packaged them up and sent them to Wall Street to sell as securitized products.

4. Types of Risks

Each financial product has its own set of risks. Some of them we already considered in early lessons while others are new. In any case, we can use this as a review. Consider, for example, that you have a choice between depositing money in a savings account, such as a certificate of deposit, or purchasing a bond. As such, you want to know what the risks are. Let's consider four different risks.

4.1 Market Risk

Market risk is the risk due to price fluctuation, which makes the value of assets go down. If you have a savings account, then you are immune to market risk because you have a specified interest rate. However, if you own a bond, the prices do change. If you bought a bond and then try to sell it, you might find that you have to sell at a lower price. As a result, you would lose money due to market fluctuation. One safe way to avoid market risk in a bond is simply to hold the bond to maturity. If you hold a bond to maturity, then the bond price can change. So long as there is no default, you are immune to price fluctuations because you never sell the bond.

4.2 Reinvestment Risk

If you decide to go with the bond, then periodically, you will receive coupon interest. You may decide not to spend that coupon interest but rather to reinvest it. What will interest rates be like when you do sell? If interest rates go up, then that means you will be able to reinvest at a higher rate of interest, which is good. However, if interest rates go down, then you would reinvest at a lower rate of interest. The risk is there. **Reinvestment risk** is the risk that rates are lower when you have access to new investment funds. Reinvestment risk exists for CDs because, upon maturity, as you seek to renew a certificate of deposit, the rate may be lower than what you originally had.

4.3 Default Risk

Default risk is the risk that your bond issue fails to pay you back the stated interest and/or the principal. As we have seen, the certificate of deposit is insured by our trustworthy depository insurance corporation, so default risk is a non-issue. If you were the holder of a risk-free bond, then likewise, you need not worry about the default risk of those securities. However, if you are the buyer and/or holder of a risky bond, then you are certainly subject to the default risk of the bond issuer. If the bond issuer defaults, then the bond will be worth less. Even in the absence of a default, the fact that the bond issuer's credit rating declines causes the bond to be worth less as the bond is perceived as riskier than before.

Again, let's look at the difference between holding the bond to maturity versus trading the bond. If you simply hold the bond to maturity, then you run the risk that there may be a default. This means that you lose the interest and principal. If you only hold the bond for a brief period of time, then you are still subject to the credit risk. The decline in credit will cause a market drop, so although it is not default risk specifically, it is credit risk in general. Default risk is just an extreme form of credit risk. Any lower credit rating results in a price drop of the bond. As a result, when you go to sell the bond, you will undoubtedly get less due to that higher credit risk. Default risk is just one type—the main type—of credit risk. Credit risk really refers to the degradation of credit worthiness. At a small scale, this results in a simple lowering of the credit score. At an extreme scale, this is actual default.

4.4 Inflation Risk

Inflation risk is the risk that the money you receive is simply worth less because inflation was higher than expected. For example, suppose either the certificate of deposit or the bond earned you 3% interest over the year. But what if inflation were 8% that year? Indeed, your real rate of return would be -5 percent. That is, had you gone to the marketplace, you could have purchased goods and services that now, by delaying a year, cost you more money; even though you have 3% more, those goods and services cost 8% more. In fact, people often find themselves in this situation in high-inflation environments. This is where central banks can help. When central banks find inflation to be high, they look to perform monetary policy, like raising interest rates, that will help to keep inflation under control.

4.5 Credit Risk Revisited

Lenders must consider several key types of risk when they originate loans:

- fraud risk;
- default risk;
- credit spread risk; and
- concentration risk

Altogether, these 4 risks are referred to as credit risk for banks or lenders in general.

Fraud risk has to do with the verification of the person or company that applies for a loan in order to distinguish legitimate applications from those that are fraudulent. Lenders need to check if the borrower is serious about repaying their debt. For instance, a client may ask to take out a loan in their name without any intention to repay it. Instead, the client may use the loan for an exotic vacation. Some would say that Countrywide Financial did not care about this at all as they often extended credit to individuals who could not really afford it. The following optional reading [Owner-Occupancy Fraud and Mortgage Performance](#) could be of interest for some students as it highlights instances of fraud.

Default risk was explained above in detail. It is worth noting that default risk is connected to **counterparty risk**. In some complicated transactions, the parties involved promise to exchange certain cash flows or exchange cash for assets. If one of the parties does not fulfill its obligations under the contract, it could lead to financial disruption for the other party. In turn, the latter party could default on its obligation toward a different third party, and so on. Therefore, lenders need to ascertain whether all parties involved can be counted on to fulfill their contractual obligations and/or need to decide how much credit to extend to the parties, if any.

Credit spread risk. Commercial bonds and risk-free government bonds or Treasury bills have a credit spread that depends on perceived levels of risk. If the spread increases, the credit risk for the borrower increases. The market value of the assets could decrease, which would result in losses for lenders and/or investors.

Concentration risk. This type of risk should be intuitive. When a large portion of a bank's portfolio is concentrated on a single sector or asset class (think of Countrywide Financial and the fact that almost their entire balance sheet was tied to the real estate industry) and when adverse events affect that sector or asset class, this concentration can cause substantial losses. Some argue that concentration could also contribute to systemic risk as the failure in one sector could cause failures in others, resulting in a domino effect.

5. Types of Marketplaces

We have already discussed different types of maturities, products, and risks, so now, we'll close this lesson by showing that there are distinct types of marketplaces as well. Broadly speaking, there are two kinds. The first type of marketplace is an **exchange**. The second is what is called **over the counter**, abbreviated as **OTC**.

Exchanges have the following characteristics:

- They offer a variety of benefits that OTC markets do not have. The main advantages of exchanges are efficiency, transparency, equitability, and credit risk mitigation.
- They primarily provide a centralized marketplace, making trading efficient. They have very well-defined rules for handling the processing and clearing of trades.
- They provide transparency because all market participants have access to the order screens that contain the bid sizes and amounts, the offer sizes and amounts, and a record of the previous trades that took place.
- The transparency and availability of data makes trading fairer for market participants. Of course, big institutional investors may have bigger infrastructure with fast computers and smart algorithms to detect and process orders more efficiently than smaller individuals or institutions do.
- They solve the problem of credit risk. When you perform a transaction on an exchange, all the credit risk is eliminated. How? The credit risk is taken by the exchange itself. In other words, suppose someone named Mamadou borrowed a security from you in order to sell it short. Soon after Mamadou sells it, the price jumps up! Now, you have more credit risk because it just became more difficult for Mamadou to cover that short. The exchange can require Mamadou to post additional financing. Financing was one of the two challenges in this module (with the other being credit risk). The exchange continually monitors the value of that security and requires Mamadou to post additional funds, even as often as intraday. In doing so, you as the lender of the security would be assured that, even with price increases, you will be able to receive the security back because Mamadou cannot run out of money to cover it. The exchange enforces this by mandating intraday requirements such that Mamadou must post cash in an account to make sure he has sufficient funds available to cover that short. If Mamadou were not able to do so, the exchange could force the sale to immediately cover the security and return it to the lender. So, the exchange's monitoring and enforcement of margin helps to ensure that financing remains creditworthy.

Ideally, all securities would be exchange traded for the advantages of equitability, transparency, and the minimization of credit risk. When we turn to OTC markets, we see that trading occurs directly between counterparties. Yes, there may be brokers in between, but the brokers do not take credit risk. The brokers are there to facilitate and manage the transaction, not to assume the credit risk of either side. Likewise, the transparency is more difficult because OTC markets decentralize the process. For example, stocks that trade on a stock exchange only trade in that one place. However, bonds may trade OTC for which there are different OTC markets. That means that selected OTC markets can have more market share than others. As a result, there could be slightly different prices from time to time between markets. Therefore, it is much harder to assure that you have received a fair price.

6. Conclusion

We started off this lesson by introducing central banks. For now, we want to emphasize that banks help to address the challenge we've seen in this module: credit risk. Credit risk appears in several places; in this module, we discussed the challenge of financing. We concluded by showing that the type of financial trading, namely exchanges, can also help mitigate credit risk. However, there are still many products that trade over the counter, so credit risk continues to be an ongoing concern for many financial transactions.

In the next lesson, we will switch asset classes by moving from fixed income to equities and cryptocurrencies. Those assets have much more volatility than fixed income does. Just as credit risk is a major challenge to fixed income, volatility will be a significant consideration for equities and cryptocurrencies.

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